WORKED SOLUTION

STEP I 2014, Question 4

Pure • Related Rates of Change

STEP I 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q4

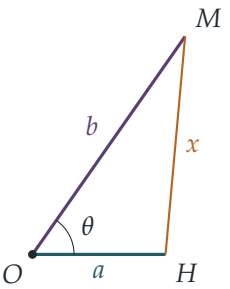
The key idea

The two hands and the line joining their tips form a triangle, so the cosine rule gives x in terms of θ. Because the clock is accurate, θ increases at a constant rate, so the rate of increase of x is that constant times $\mathrm{d}x/\mathrm{d}\theta$. Maximising this is then an ordinary one-variable calculus problem, and the stationary-point equation turns out to be a quadratic in $\cos\theta$ that factorises neatly.

Solution

The rate of increase of x

Let O be the pivot, H the tip of the hour hand and M the tip of the minute hand, so $OH = a$, $OM = b$, $\angle HOM = \theta$ and $HM = x$.



By the cosine rule,

$$x^2 = a^2 + b^2 - 2ab\cos\theta.$$

Note that $x \geq b - a > 0$, so we may divide by x freely. Differentiating with respect to time t,

$$2x\frac{\mathrm{d}x}{\mathrm{d}t} = 2ab\sin\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}t}.$$

The clock is accurate, so each hand turns at a constant angular speed, and the angle between them changes at a constant rate $\omega > 0$ (the difference of the two angular speeds; while $0 \leq \theta < \pi$ the minute hand is pulling ahead, so θ is increasing). Hence

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \omega ab \frac{\sin\theta}{x}.$$

So we must find where $f(\theta) = \dfrac{\sin\theta}{x}$ is greatest on $0 \leq \theta < \pi$, remembering that x depends on θ with $\dfrac{\mathrm{d}x}{\mathrm{d}\theta} = \dfrac{ab\sin\theta}{x}$.

Maximising

By the quotient rule,

$$f'(\theta) = \frac{x\cos\theta - \sin\theta \cdot \dfrac{ab\sin\theta}{x}}{x^2} = \frac{x^2\cos\theta - ab\sin^2\theta}{x^3}.$$

Write $c = \cos\theta$. Using $x^2 = a^2 + b^2 - 2abc$ and $\sin^2\theta = 1 - c^2$, the numerator is

$$\begin{aligned} x^2\cos\theta - ab\sin^2\theta &= (a^2+b^2)c - 2abc^2 - ab + abc^2 \\ &= -(abc^2 - (a^2+b^2)c + ab) \\ &= -(ac-b)(bc-a) \\ &= (b-ac)(bc-a). \end{aligned}$$

Ask yourself How do we know this stationary point is a maximum, and the only one?

Since $c \leq 1 < b/a$, we have $ac < b$, so the factor $b - ac$ is always positive. Also $x^3 > 0$. So $f'(\theta)$ has the same sign as $bc - a$, that is, of $\cos\theta - \dfrac{a}{b}$:

- when $\cos\theta > a/b$ (smaller θ), $f'(\theta) > 0$ and f is increasing;
- when $\cos\theta < a/b$ (larger θ), $f'(\theta) < 0$ and f is decreasing.

As $\cos\theta$ is decreasing on $[0, \pi)$ and $0 < a/b < 1$, there is exactly one θ in the range with $\cos\theta = a/b$, and f rises up to it and falls after it. So this gives the global maximum of the rate of increase of x. There,

$$x^2 = a^2 + b^2 - 2ab \cdot \frac{a}{b} = b^2 - a^2,$$

so the rate of increase of x is greatest when $x = (b^2 - a^2)^{1/2}$, as required.

Geometrically, $x^2 + a^2 = b^2$ says the triangle has a right angle at H: the rate is greatest when the line joining the tips is perpendicular to the hour hand.

The case $b = 2a$

Now $\cos\theta = a/b = \frac{1}{2}$, so the maximum rate occurs when $\theta = \pi/3$.

Measure t in minutes after mid-day. The minute hand turns through 2π in 60 minutes and the hour hand through 2π in 720 minutes, both clockwise from the vertical. So the angle between them is

$$\theta = 2\pi\left(\frac{1}{60} - \frac{1}{720}\right)t = 2\pi \cdot \frac{11}{720}\,t = \frac{11\pi}{360}\,t,$$

which is valid while this is less than π. Starting from $\theta = 0$ at mid-day, θ first reaches $\pi/3$ when

$$\frac{11\pi}{360}\,t = \frac{\pi}{3} \quad\Longrightarrow\quad t = \frac{120}{11} = 10\tfrac{10}{11} \text{ minutes.}$$

This is about 10.91 minutes (10 minutes and roughly 55 seconds), a little less than 11 minutes, as required.

Checks

- At $\theta = 0$ the rate is 0 (the hands are together and $x = b - a$ is at its minimum), and as $\theta \to \pi$ it tends to 0 again (where $x \to a + b$ is at its maximum). So a maximum strictly in between is exactly what we should expect.
- With $a = 1$, $b = 2$, tracking the positions of the hand tips minute by minute and estimating the rate of change of their distance, the greatest rate appears at $t \approx 10.909$ minutes, where $x \approx 1.732 = \sqrt{3} = \sqrt{b^2 - a^2}$.
- The answer is just under 11 because if the hour hand stood still the minute hand would need exactly 10 minutes to reach 60°; the hour hand creeps forward, so a little extra time is needed, and $120/11$ is the same kind of number as the familiar "the hands next coincide after $720/11$ minutes".

Common slips

- Maximising x itself, or $\mathrm{d}x/\mathrm{d}\theta$ without noticing that x also depends on θ in the quotient-rule step.
- Using the minute hand's speed alone ($2\pi/60$ per minute) for $\mathrm{d}\theta/\mathrm{d}t$, which gives exactly 10 minutes; the angle between the hands grows at the *difference* of the two speeds.
- Forgetting to discard the root $\cos\theta = b/a$, which is impossible because $b > a$.

Beyond STEP

This is a related-rates problem: one quantity changes at a known rate and we track how a geometrically linked quantity responds. The same idea, describing positions

by angles that vary uniformly in time, is the starting point for planetary motion: the distance between two planets on circular orbits behaves exactly like x here, and the moments of fastest separation are found the same way. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this kind of kinematics with vectors.

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